Week 2

Annotations

De Morgan's Laws Proof

① $\left(\bigcup_n S_n\right)^c = \bigcap_n S_n^c$

$(S_1 \cup S_2)^c = S_1^c \cap S_2^c$

$x \in (S_1 \cup S_2)^c \iff x \in S_1^c \cap S_2^c$

Forward Direction ($\Rightarrow$):

let $x \in (S_1 \cup S_2)^c$

$x \notin S_1 \cup S_2$

$x \notin S_1$ AND $x \notin S_2$

$x \in S_1^c$ AND $x \in S_2^c \Rightarrow x \in S_1^c \cap S_2^c$

De Morgan's Laws Proof

(2) $\left(\bigcap_n S_n\right)^c = \bigcup_n S_n^c$

$(S_1 \cap S_2)^c = S_1^c \cup S_2^c$

$x \in (S_1 \cap S_2)^c \iff x \in S_1^c \cup S_2^c$

Reverse Direction ($\Leftarrow$)

Let $x \in S_1^c \cup S_2^c$

$x \in S_1^c$ OR $x \in S_2^c$

$x \notin S_1$ OR $x \notin S_2$

$x \notin S_1 \cap S_2 \implies x \in (S_1 \cap S_2)^c$

Prove that the Probability for Event $A \leq 1$



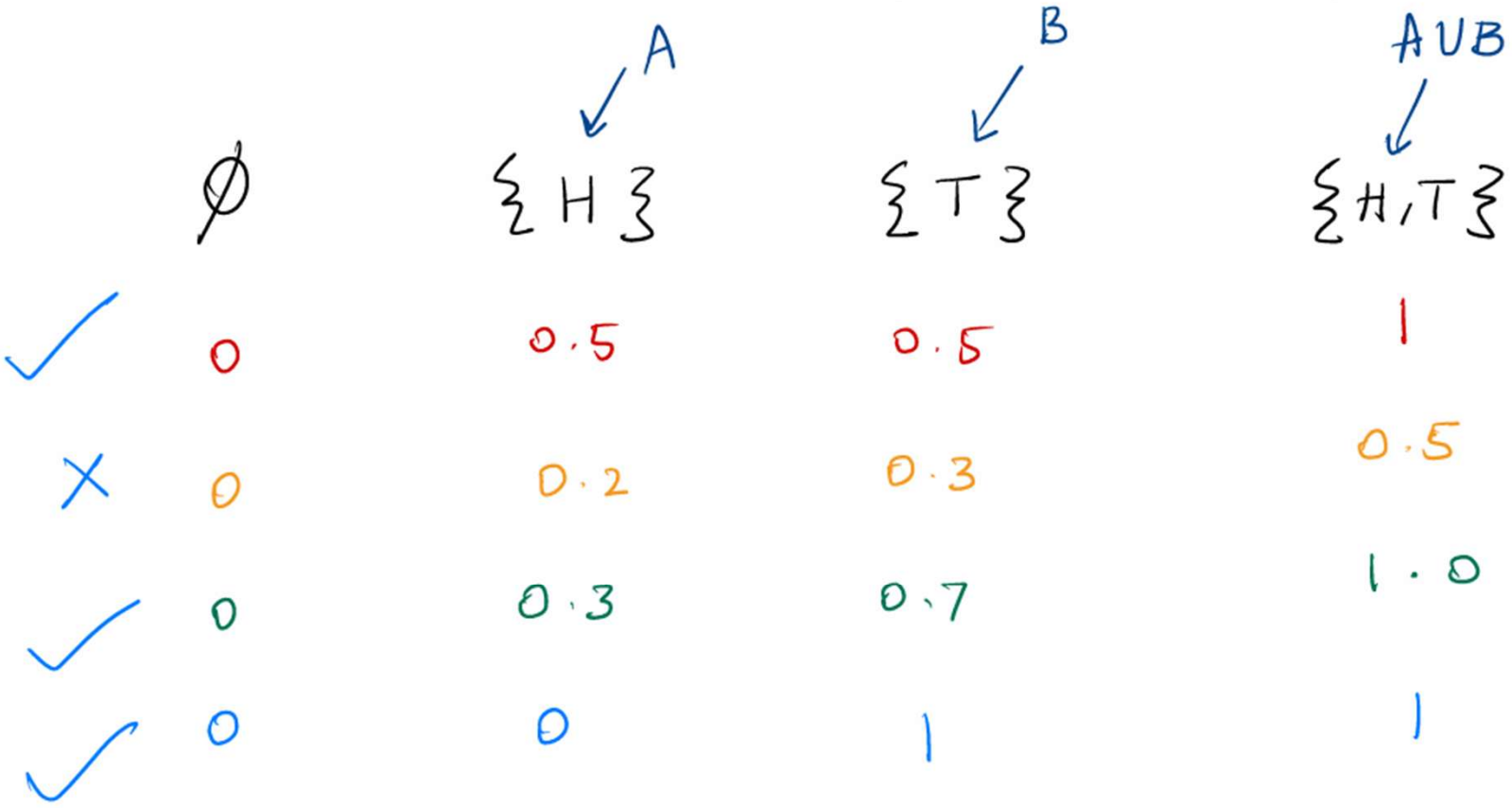
$$1 \overset{(2)}{=} P(\Omega) = P(A \cup A^c)$$

$$\overset{(3)}{=} P(A) + P(A^c)$$

$$P(A) = 1 - P(A^c) \overset{(1)}{\leq} 1$$

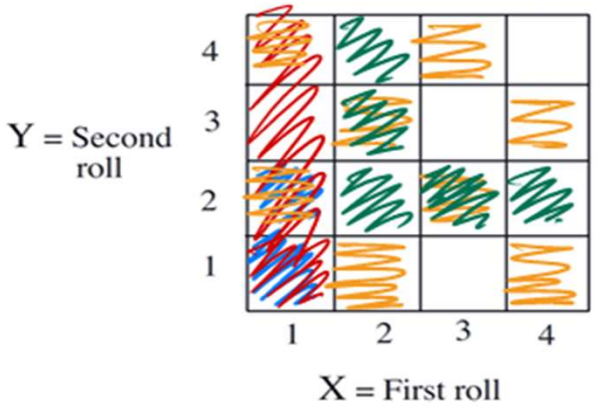
$$P(A^c) \geq 0$$

For a coin toss, which of the following are valid Probability Laws?



Probability Law: Example with finite sample space

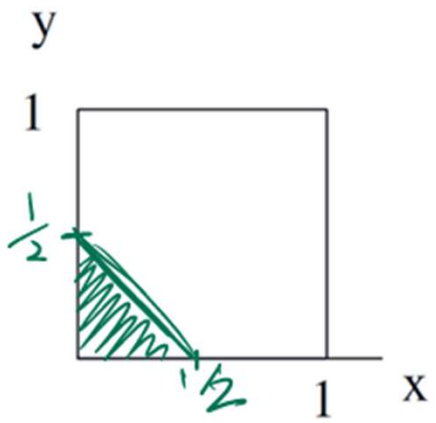
Fair set of dies



- Let every possible outcome have probability 1/16
  - $\mathbf{P}((X,Y) \text{ is } (1,1) \text{ or } (1,2)) =$ $2/16 = 1/8$
  - $\mathbf{P}(\{X = 1\}) =$ $4/16 = 1/4$
  - $\mathbf{P}(X + Y \text{ is odd}) =$ $8/16 = 1/2$
  - $\mathbf{P}(\min(X,Y) = 2) =$ $5/16$

Continuous Uniform Law

- Two "random" numbers in $[0,1]$.



- Uniform law: Probability = Area
  - $\mathbf{P}\left(X+Y \leq \frac{1}{2}\right) = \frac{1}{8} \div 1 = \frac{1}{8}$
  - $\mathbf{P}\big((X,Y) = (0.5, 0.3)\big) = 0$

Some consequences of the axioms...

For disjoint events A, B and C, what is $\mathbf{P}(A \cup B \cup C)$

$\overset{(3)}{=} P(A \cup B) + P(C)$

$\overset{(3)}{=} P(A) + P(B) + P(C)$

$P(S_1 \cup S_2 \cup \cdots \cup S_{1000})$

$= P(S_1) + P(S_2) + \cdots + P(S_{1000})$

Some consequences of the axioms…

$\mathbf{P}(\emptyset) = ?$

$\Omega^c = \emptyset$

$\Omega \cup \emptyset = \Omega$

$P(\Omega \cup \Omega^c)$

$\Rightarrow P(\Omega \cup \emptyset) = P(\Omega) + P(\emptyset)$

$\Rightarrow P(\Omega) = P(\Omega) + P(\emptyset)$

$\Rightarrow 1 = 1 + P(\emptyset)$

$P(\emptyset) = 0$

Choosing the Right Sample Space

Example 1.1. Consider two alternative games, both involving ten successive coin tosses:

Game 1: We receive \$1 each time a head comes up.

Game 2: We receive \$1 for every coin toss, up to and including the first time a head comes up. Then, we receive \$2 for every coin toss, up to the second time a head comes up. More generally, the dollar amount per toss is doubled each time a head comes up.

What is an appropriate sample space for each of these experiments?

$\{0, 1, 2, 3, \ldots, 10\}$

2^{10} or 1024 possibilities

$\{HHH\cdots H, H\cdots HT, \ldots, TTT\cdots T\}$

Utilizing Probability Theory for Dealing with Uncertainty: Two Stages

1. Understand what your experiment is

   "Tossing a coin" Is it fair? Yes.

2. Construct a probabilistic model
   - Specify the sample space $\Omega = \{H, T\}$
   - Specify the probability law (implicitly/indirectly)

   $P(H) = 1/2$, $P(T) = 1/2$

Probabilistic Modelling

3. Utilize the probabilistic model to derive the probabilities of certain events of interest
   - Identify an event of interest
   - Calculate its probability

Probabilistic Reasoning

A Simple Example

Tim has a four-sided dice. When he rolls the dice, all the possible outcomes (1, 2, 3, 4) are equally likely.

What is the probability of outcome being even?

1. Exp: "Rolling a fair 4-sided dice"
2. $\Omega = \{1, 2, 3, 4\}$
3. $P(1) = 1/4$, $P(2) = P(3) = P(4) = 1/4$
4. Event of Inter.: $E = \{2, 4\}$
5. Calculate Probabilities: $P(E) = P(2) + P(4)$
   $= 1/4 + 1/4 = 1/2$

A slightly more complicated example...

Tim has a peculiar four-sided dice. When he rolls the dice, the probability of getting any particular outcome is proportional to result of the dice.

What is the probability of outcome being even?

1. Exp: "Rolling an unfair 4-sided die"
2. $\Omega = \{1, 2, 3, 4\}$
3. $P(1) = 1/10$, $P(2) = 2/10$, $P(3) = 3/10$, $P(4) = 4/10$
4. Event of Interest
5. $E = \{2, 4\}$

$P(E) = 2/10 + 4/10 = 3/5$

let $x = P(1)$

$P(2) = 2x$, $P(3) = 3x$

$P(4) = 4x$.

$P(\Omega) = 1$

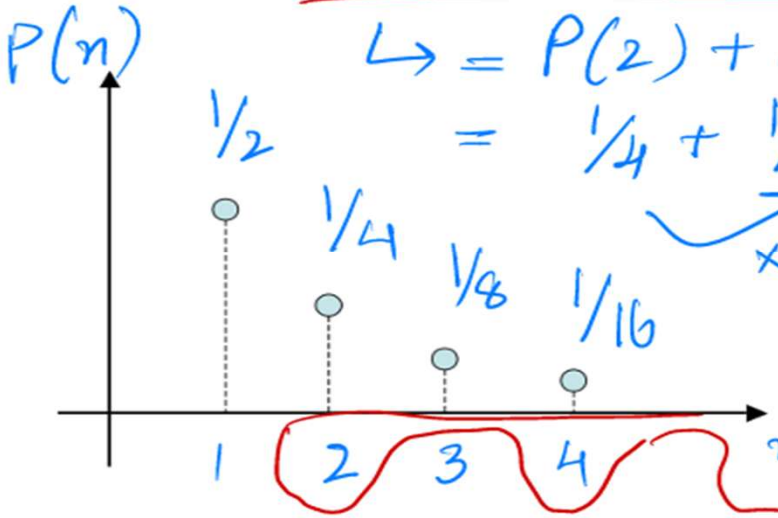
$1 = x + 2x + 3x + 4x$

$x = 1/10$

Probability Law: Example w/ Countably Infinite Sample Space

- Sample space: $\{1, 2, 3, \ldots\}$
  - We are given $\mathbf{P}(n) = 2^{-n}$, $n = 1, 2, 3, \ldots$
  - Find $\mathbf{P}$(outcome is even)

EXP: "Flip a coin until I get a H. Then I stop"



$\hookrightarrow = P(2) + P(4) + P(6) + \cdots$

$= 1/4 + 1/16 + 1/64 + \cdots$

$\times 1/4$

$= \sum_{k=0}^{\infty} (1/4) \cdot (1/4)^k$

$= \frac{1/4}{1 - 1/4}$

$= 1/3$

Example 3

Out of the students in a class, 60% love Cricket, 70% love Soccer, and 40% love both Cricket and Soccer. What is the probability that a randomly selected student loves neither Cricket nor Soccer.

1. Exp: "Selecting a Random Student"
2. $\Omega = \{CS, CS^c, C^cS, C^cS^c\}$ ✓✓
3. $P(CS) = 0.4$
4. $P(CS \cup CS^c) = 0.6$
5. $P(CS) + P(CS^c) = 0.6$

$P(CS^c) = 0.6 - 0.4 = 0.2$

Example 3

Out of the students in a class, 60% love Cricket, 70% love Soccer, and 40% love both Cricket and Soccer. What is the probability that a randomly selected student loves neither Cricket nor Soccer.

1. $P(CS \cup C^cS) = P(CS) + P(C^cS)$
2. $0.7 = 0.4 + P(C^cS)$
3. $P(C^cS) = 0.3$
4. $P(\Omega) = P(CS) + P(C^cS) + P(CS^c) + P(C^cS^c)$
5. $1 = 0.4 + 0.3 + 0.2 + P(C^cS^c)$

$P(C^cS^c) = 0.1$